# Robustness Verification for Object Detectors using Set-Based Reachability Analysis

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# Object Detection

Image Classifiers: $f(I) = c$
Object Detectors: $f(I) = \{O_1, O_2, \ldots, O_i, \ldots O_k\}$

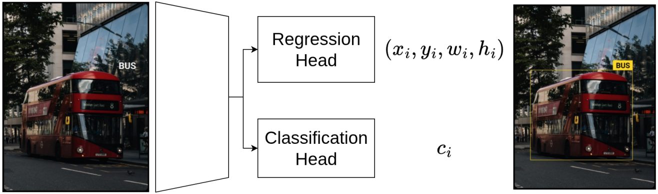



Figure: An Object Detectors with Regression and Classification Head.

where each $O_i$ is a tuple $(x_i, y_i, w_i, h_i, c_i)$, containing bounding box coordinates $x_i, y_i$, dimensions $w_i, h_i$ and the class label of the object in the image $c_i$.

# Adversarial Attacks

$$x' = x + \alpha\eta$$

$\eta$ is the adversarial noise added to original input image $x$, such that Object Detector deviates from predicted behaviour, $\alpha$ states a bound on the adversarial noise.

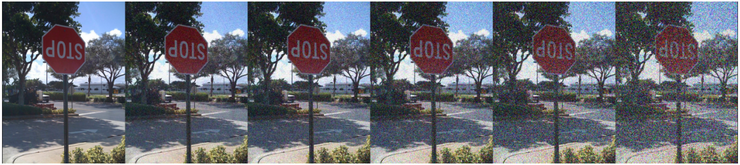



Figure: Gaussian noise applied to image $I$ from severity 1 to 5.

# Object Detectors under Adversarial Attack

Ground Truth and Detections:

$$f(I) = \{O_1, O_2, \ldots, O_i, \ldots O_k\}$$
$$\mathbf{G} = \{G_i\}_{i=1}^{k}, \quad G_i = (x_i, y_i, w_i, h_i, c_i)$$

$$f(I') = \{\hat{O}_1, \hat{O}_2, \ldots, \hat{O}_j, \ldots, \hat{O}_{k'}\}$$
$$\mathbf{D} = \{D_j\}_{j=1}^{k'}, \quad D_j = (\hat{x}_j, \hat{y}_j, \hat{w}_j, \hat{h}_j, \hat{c}_j)$$

As per the goal of the adversarial example $I' = I + \alpha\eta$, the model behaves differently under the original image $I$ and perturbed variation of the image $I'$.

Intersection over Union: $\mathrm{IoU}(G_i, D_j) = \frac{|G_i \cap D_j|}{|G_i \cap D_j|}$

# Taxonomy of Errors

**Misclassification** for some $G_i, D_j$, predicted label does not match ground truth, s.t., $\text{IoU}(G_i, D_j) \geq \tau$ but $c_i \neq \hat{c}_j$, where $\tau$ is some threshold lower bound tolerance.

**Mislocalization** for a pair $G_i, D_j$ with $c_i = \hat{c}_j$, $\text{IoU}(G_i, D_j) < \tau$, where $\tau$ is some threshold lower bound tolerance.

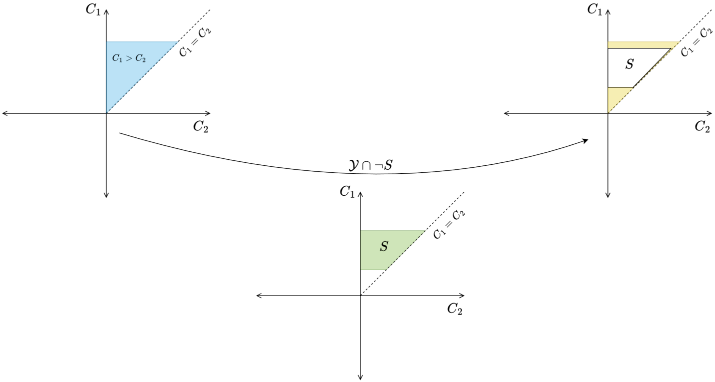
**Misdetection** missed or spurious objects leading to $k \neq k'$

# Verification of Robustness

For a model $f : \mathbb{R}^n \to \mathbb{R}^m$, an input set $\mathcal{X} \subseteq \mathbb{R}^n$ and a safety property $S \subseteq \mathbb{R}^m$, Verification involves checking whether

$$\forall x \in \mathcal{X}, f(x) \in S$$

and identifying a counterexample $x^* \in \mathcal{X}$ such that $f(x^*) \notin S$.



# Imagestars

$$\Theta = \langle c, V, P \rangle$$

$$c \in \mathbb{R}^{h \times w \times nc} \quad V = \{v_1, v_2, \ldots, v_m\} \quad P : \mathbb{R}^m \to \{\top, \bot\}$$

Imagestar is the set of reachable states from the state $x$

$$[\![\Theta]\!] = \{x \mid x = c + \sum_{i=1}^{m}(\alpha_i v_i), \ \text{s.t.} \ \ P(\alpha_1, \ldots, \alpha_m) = \top\}$$

Linear predicates have been used such that $P(\alpha) \equiv A\alpha \leq b$. Input set defined on the image with some adversarial pattern:

$$\Theta_{\mathcal{I}} = \{\hat{I} \mid \hat{I} = I + \sum_{i=1}^{n} \alpha_i V_i, \ P_{\mathcal{I}}(\vec{\alpha}) = \top\}$$

# Verifying Reachable Set

We propagate $\Theta_{\mathcal{I}}$ through the network, layer-wise, to obtain reachable outputs. Robustness is reduced to an emptiness check. Reachable output set: $\Theta_{\mathcal{O}}$
*Safety region* $S$ encodes all acceptable outputs; robustness iff

$$\Theta_{\mathcal{O}} \cap \neg S = \phi$$

.

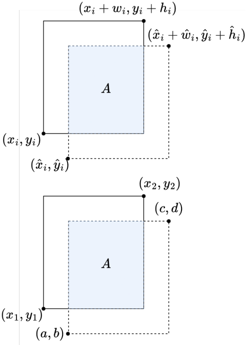
For classification, for target class $i$, $S \triangleq \bigwedge_{j \neq i} (C_j - C_i \geq 0)$. The constraints can be stacked as:

$$\begin{bmatrix} 1 & 0 & 0 & \dots & -1 & \dots & 0 \\ 0 & 1 & 0 & \dots & -1 & \dots & 0 \\ & & & \dots & -1 & \dots & 0 \\ 0 & 0 & 0 & \dots & -1 & \dots & 1 \end{bmatrix}_{(n-1)\times n} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}_{n\times 1} \leq \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(n-1)\times 1}$$

# Coordinate Transforms

Ground-truth corners $x_1 = x_i$,
$y_1 = y_i$ and $x_2 = (x_i + w_i)$, $y_2 = (y_i + h_i)$

Similarly,
predicted corners as variables $a = \hat{x}_j$,
$c = (\hat{x}_j + \hat{w}_j)$, $b = \hat{y}_j$ and $d = (\hat{y}_j + \hat{h}_j)$
for any specific predicted object $D_j$.



# Object Detection Constraints

Horizontal Overlap Constraint:

$$\min(y_2, d) - \max(y_1, b) \geq \epsilon_W \times (y_2 - y_1), \quad \epsilon_W \in [0, 1]$$

Vertical Overlap Constraint:

$$\min(x_2, c) - \max(x_1, a) \geq \epsilon_H \times (x_2 - x_1), \quad \epsilon_H \in [0, 1]$$

The resulting conjunction of linear inequalities $L_{D_j}(\alpha)$ represents the constraints for verifying the localization of a single object detection:

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} \max(x_1, a) - \min(x_2, c) \\ \max(y_1, b) - \min(y_2, d) \end{pmatrix} \leq \begin{pmatrix} \epsilon_W \times (x_2 - x_1) \\ \epsilon_H \times (y_2 - y_1) \end{pmatrix}$$

# Algorithm

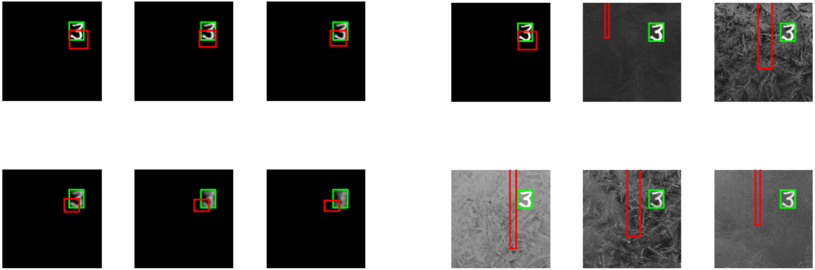
**Algorithm 1** Robustness Verification of Object Detectors

**Input:** Object Detector $f$, Input Image $I$, Perturbed Image $I'$, $\epsilon_W, \epsilon_H$
**Result:** $r : \{r_1, r_2, \ldots, r_{k'}\}$

```
r ← {}
D ← f(I)                                                ▷ D : {D_1, D_2, ..., D_k'}
Θ_I ← construct_imagestar(I, I')
Θ_O ← reach(f, Θ_I)
for D_i ∈ D do
    [S_i, C_i, B_i] ← D_i        ▷ S_i: Score, C_i: Class, B_i: Bounding Box [x, y, w, h]
    r_c ← verify_classification(Θ_O, C, S)
    r_l ← verify_localization(Θ_O, B, ε_W, ε_H)
    r_i ← (r_c, r_l)
    r ← r ∪ {r_i}
end for
return r
```

Figure: For a single detected object there are constraints $S_{D_j} = C_{D_j} \wedge L_{D_j}$

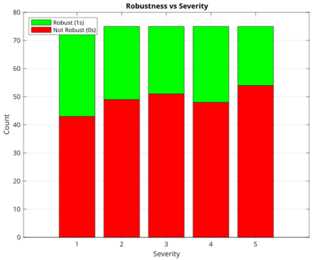
# Results



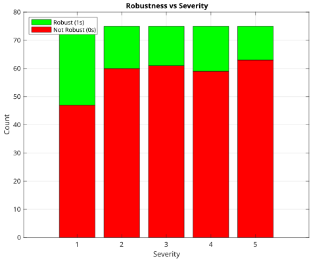
(a) `defocus_blur` (b) `frost`

Figure: Bounding box accuracy gets affected with increasing severity of corruption types `defocus_blur` (left) and `frost` (right).

# Results




(a) $\epsilon_W = \epsilon_H = 0.6$




(b) $\epsilon_W = \epsilon_H = 0.8$

Figure: Total number of Robust (green) and Non-Robust detections (red).

# Conclusions

- Safety can be defined as the conjunction of formalized classification and localization constraints, capturing misclassification, mislocalization, and misdetection errors.
- ImageStars can be extended with new constraints to formally verify both classification and localization in object detectors.
- For each object we will have $S_{D_j} = C_{D_j} \wedge L_{D_j}$, thus for all detected objects in an image we have $S = \bigwedge_{i=1}^{k'} S_{D_j}$.